reasoning nor spell out his further, more detailed, assumptions. Instead I shall go directly to the conclusion. He did indeed find an evolutionarily stable pair of rules.

So, to the big question. Does Grafen's ESS constitute the kind of world that Zahavi would recognize as a world of handicaps and honesty? The answer is yes. Grafen found that there can indeed be an evolutionarily stable world that combines the following Zahavian properties:

1. Despite having a free strategic choice of advertising level, males choose a level that correctly displays their true quality, even if this amounts to betraying that their true quality is low. At ESS, in other words, males are honest.

2. Despite having a free strategic choice of response to male advertisement, females end up choosing the strategy 'Believe the males'. At ESS, females are justifiably 'trusting'.

3. Advertising is costly. In other words, if we could somehow ignore the effects of quality and attractiveness, a male would be better off not advertising (thereby saving energy or being less conspicuous to predators). Not only is advertising costly; it is because of its costliness that a given advertising system is chosen. An advertising system is chosen precisely because it actually has the effect of reducing the success of the advertiser—all other things being held equal.

4. Advertising is more costly to worse males. The same level of advertising increases the risk for a puny male more than for a strong male. Low-quality males incur a more serious risk from costly advertising than high-quality males.

These properties, especially 3, are full-bloodedly Zahavian. Grafen's demonstration that they are evolutionarily stable under plausible conditions seems very convincing. But so also did the reasoning of Zahavi's critics who influenced the first edition of this book and who concluded that Zahavi's ideas could not work in evolution. We should not be happy with Grafen's conclusions until we have satisfied ourselves that we understand where—if anywhere—those earlier critics went wrong. What did they assume that led them to a different conclusion? Part of the answer seems to be that they did not allow their hypothetical animals a choice from a continuous range of strategies. This often meant that they were interpreting Zahavi's verbal ideas in one or other of the first three kinds of interpretation listed by Grafen—the Qualifying Handicap, the Revealing Handicap, or the Conditional Handicap. They did not consider any version of the fourth interpretation, the Strategic Choice Handicap. The result was either that they couldn't make the handicap principle work at all or that it worked but only under special, mathematically abstract conditions, which did not have the full Zahavian paradoxical feel to them. Moreover, an essential feature of the Strategic Choice interpretation of the handicap principle is that at ESS high-quality individuals and low-quality individuals are all playing the same strategy: 'Advertise honestly'. Earlier modellers assumed that high-quality males played different strategies from low-quality males, and hence developed different advertisements. Grafen, in contrast, assumes that, at ESS, differences between high- and low-quality signallers emerge because they are all playing the same strategy—and the differences in their advertisements emerge because their differences in quality are being faithfully rendered by the signalling rule.

We always admitted that signals as a matter of fact can be handicaps. We always understood that extreme handicaps could evolve, especially as a result of sexual selection, *in spite of* the fact that they were handicaps. The part of the Zahavi theory that we all objected to was the idea that signals might be favoured by selection precisely *because* they were handicaps to the signallers. It is this that Alan Grafen has apparently vindicated.

If Grafen is correct—and I think he is—it is a result of considerable importance for the whole study of animal signals. It might even necessitate a radical change in our entire outlook on the evolution of behaviour, a radical change in our view of many of the issues discussed in this book. Sexual advertisement is only one kind of advertisement. The Zahavi–Grafen theory, if true, will